16TH NOTE

XBOX DRIVER CONTROLS

Connect one Xbox controller to Driver Station USB 0. This is the only supported controller layout.

Driver controls

Drive forward / reverse	Left stick Y
Turn left / right	Right stick X
Lower intake and collect	Hold LB
Stop collecting; leave intake down	Release LB
Shoot: spin up, raise intake, then feed	Hold RB
Stop shooting; return intake down	Release RB
Request existing game-piece vision assist	Hold RT over 50%

How the controls behave

INTAKE	The roller starts only after the intake reaches the deployed position. Releasing LB stops the roller but leaves the intake down.
SHOOT	Both flywheels must be at speed before the intake raises and the indexer feeds. A speed drop stops feeding immediately.
LB + RB	Shooting wins. Intake collection stays off while RB is held.
VISION	Manual stick movement over 0.15 cancels assist. Invalid or stale vision returns control to the sticks. This is the existing game-piece assist, not PhotonVision AprilTag tracking.
ARMING AND SAFETY	After enable, controller reconnect, or command interruption, release LB and RB before requesting intake or shooting. Disable, faults, timeouts, or controller disconnect stop commanded output. Flywheels may coast.

16TH NOTE

SETUP AND TEST CONTROLS

Use these controls only with the robot supported and the mechanism area clear.

Disabled dashboard actions

Fuel/Apply configuration (disabled)	Disabled	Stops fuel outputs, applies staged motor/control settings, and invalidates the intake reference. Reference again after every full apply.
Fuel/Confirm intake at upper stop	Disabled	With the intake physically supported at the upper frame-contact stop, zeros and verifies the pivot encoder. It never moves the motor.

Test-mode commissioning

1	Fuel/Test/Mechanism	Select PIVOT, INDEXER_PAIR, or SHOOTERS while disabled.
2	Fuel/Test/Volts	Enter the intended signed test voltage. Start low. Output is clamped by the configured test and normal voltage caps.
3	Driver Station	Enable Test mode. The drivetrain is held stopped.
4	Xbox RB	Release first, then hold RB as the deadman control. Release it to stop.

Maximum continuous hold before release/rearm: pivot 0.25 s; indexer pair 1 s; shooter pair 5 s. Pivot testing requires a valid reference. Shooter testing requires a confirmed coupling selection.

Autonomous chooser

Do nothing (default)	Continuously stops drivetrain and fuel output. Use this until autonomous is fully commissioned.
Back up and shoot preloads	Requires verified REAL fuel/auto settings, healthy devices, and a valid intake reference. Test the exact starting position before use.

AdvantageScope tuning

Tuning - Fuel - FeedDegrees	Sets the normal raised shooting/feed position. Increase the value to stop the intake lower.
Tuning - Fuel - AgitationDegrees	Sets the intermediate agitation position. Agitation remains optional and should stay disabled until normal feeding is reliable.
Fuel/AngleTuningStatus	Reports whether the edit was accepted, rejected, or waiting for the real robot to be disabled.

QUICK STATUS CHECK

Before normal fuel operation, require Fuel/DevicesConfigured = true, Fuel/Referenced = true, Fuel/AutomaticReady = true, and no Fuel/Fault message. REAL dashboard device settings are temporary until copied into FuelConstants.REAL.